

Milken, the “Junk Bond King” (who went to prison for securities fraud). Prior to this we could not access a high yield, or junk, credit risk premium. Finally, we could only collect the carry factor in foreign exchange markets when the world’s currencies became untethered from gold and freely floated in 1970s.¹⁴

There is no consensus on the complete set of factors. Academic studies using statistical analysis suggest that a limited number, usually less than ten, can capture most variation in expected returns.¹⁵ These are, however, statistical and not economic descriptions of data. Some factors, like CPPIB’s bond and equities, and some dynamic factors like value-growth and momentum, are tried and true. I recommend that even if you can confidently identify and invest in ten or more factors, don’t. Keep things simple. Start with a select few. The point is that putting a factor into a fund’s benchmark removes a component from the active return that doesn’t really belong there and can be obtained more cheaply.

Some factors disappear. The *size effect*—that small stocks outperform large stocks—was brought to investors’ attention by Banz in 1981 and reached its peak just after that. There are good academic justifications why small stocks might do better than large stocks: smaller stocks are more illiquid, they are less likely to be followed by analysts, so information is scarcer, small stocks don’t have the same operating cushions that large firms do, and they operate in riskier segments of the economy. Since the mid-1980s, however, there has been no size premium after adjusting for market risk (see chapter 7). That is, small stocks do have higher returns than large stocks but not after taking out their exposure to the market factor. The creation of small stock mutual funds allowed the ordinary investor to participate and bear size-related risk. Thus the risk-bearing capacity of the economy changed after industry created new products to capitalize on the size premium. Those industry developments caused size to disappear. Those early investors in small caps experienced a bonanza. Prices of small stocks rose back to long-run equilibrium and early investors enjoyed a tidy risk-adjusted capital gain.

As I write, there is a flurry of new products created by industry to exploit many factor risk premiums. I take a special interest in funds taking advantage of the high average returns to low risk stocks, or *the risk anomaly*, which I describe in chapter 10. Stocks with low volatilities, or low (past) betas, seem to be too cheap as they have anomalously high returns. Low volatility could be a factor, according to the criteria I lay out, but admittedly the academic literature on the risk anomaly is still much smaller than the literature on value-growth, momentum, and selling volatility protection. As the asset management industry creates products to trade low volatility risk—as it has been doing with gusto (although perhaps not as cheaply as asset owners would wish)—new investors may increase their holdings of low risk stocks. Perhaps then prices of low risk stocks will rise to where they

¹⁴ Accominotti and Chambers (2013) report that carry worked even worked during the 1920s and 1930s, and many countries were on the gold standard for some of this time.

¹⁵ See Connor and Korajczyk (1993) and Jones (2001).